

VIPUL PANDEY

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https://github.com/vipulcodez

DATA ANALYST

Computer Science & Engineering graduate with hands-on experience in Python, SQL, data analysis, machine learning, business intelligence, and software development. Experienced in building data-driven applications, predictive models, interactive dashboards, and API-based ML solutions. Strong foundation in Data Structures & Algorithms, DBMS, OOP, ETL, data visualization, and machine learning, with 800+ programming problems solved.

WORK EXPERIENCE

Centre for Railway Information Systems (CRIS) Intern

01/2025 - 05/2025

- Assisted in the development and maintenance of IT systems supporting Indian Railways' operations, contributing to modules related to passenger/freight management and data reporting.
- Gained hands-on experience with real-time data processing, system integration, and government-scale software infrastructure.

EDUCATION

Bachelor of Technology in Computer Science & Engineering

I.K. Gujral Punjab Technical University (IKGPTU) • GPA: 7.4

01/2021 - 01/2025

ACHIEVEMENTS

Solved 800+ problems on different platform including [geekforgeeks](#) and [leetcode](#). Gold Medal In Shot Put AND Silver Medal in Hammer Throw

MIMIT Annual Sports Day

PROJECTS

ML Deployment Platform | Python, Scikit-learn, FastAPI, Docker, Git/GitHub

- Built an end-to-end **customer churn prediction platform** using Python and Scikit-learn.
- Developed reusable **preprocessing, model training, and inference pipelines** for production-ready predictions.
- Exposed the trained ML model through a **FastAPI REST API** with interactive Swagger documentation.
- Containerized the application using **Docker** for reproducible deployment and environment consistency.
- Implemented real-time predictions with **churn probability scoring** and maintained the project using Git/GitHub.

Customer Churn Prediction | Python, SQL, Power BI, Scikit-learn

- Cleaned and analyzed 7,000+ customer records to prepare data for modeling.
- Built a machine learning model to predict customer churn.
- Designed an interactive Power BI dashboard to visualize churn KPIs and customer insights.
- Identified key churn drivers including contract type, payment method, and tenure.

Social Media Sentiment Analysis | Python, Pandas, NLP, Power BI.

- Processed and analyzed social media posts using NLP techniques.

- Classified sentiments into Positive, Neutral, and Negative categories.
- Created visual dashboards to track sentiment trends and audience engagement.
- Generated insights to support marketing and customer experience strategies.

Movie Recommender System | Python, Pandas, Scikit-learn, SQL.

- Engineered a recommendation engine that intelligently suggests the top 5 similar movies based on user input.
- Utilized collaborative filtering techniques and the K-Nearest Neighbors (KNN) algorithm to improve prediction accuracy.
- Leveraged NumPy and Pandas to efficiently preprocess and analyze large datasets, ensuring seamless data handling.
- Enhanced* user personalization by dynamically comparing feature vectors to generate relevant movie suggestions.

VOLUNTEERING & LEADERSHIP

Campus Ambassador — MIMIT Training & Placement Office | 2021–2024

11/2021 - 10/2024

- Coordinated student–recruiter communication and supported campus recruitment activities.
- Organized employability and placement–preparation sessions for students.
- Managed student/employer databases and tracked recruitment progress.

Volunteer — Takshila Educational Society

05/2017 - 01/2019

- Supported educational and community–focused initiatives aimed at improving learning opportunities for students.
- Assisted in organizing educational activities, awareness programs, and community outreach initiatives.
- Coordinated with volunteers and team members to ensure smooth execution of social and educational programs.
- Interacted with students and community members, developing strong communication, teamwork, and organizational skills.

PUBLICATIONS

DEEP LEARNING WITH UNSUPERVISED AND SUPERVISED APPROACHES IN MEDICAL IMAGE ANALYSIS

11/2024

[IRJMETS](#) (Impact Factor: 8.187)

- Published research on hybrid deep learning techniques for medical image analysis in IRJMETS (Impact Factor: 8.187).
- Proposed scalable solutions combining supervised and unsupervised methods for efficient disease diagnosis and segmentation.
- Highlighted applications of CNNs, GANs, and U-Net in addressing labeled data scarcity and enhancing model performance.

SKILLS

Programming: C, C++, Python, SQL

Data Analysis: Excel (Pivot Tables, VLOOKUP), NumPy, Pandas

Visualization: Matplotlib, Power BI, Seaborn, Tableau

Databases: MongoDB, MySQL

Concepts: DBMS, ETL, Machine Learning, OOPS